Viktor Stein

Postdoc, Applied Mathematics

viktor.stein@tum.de
https://viktorajstein.github.io/

Education

04/2023 - 03/2026	PhD (Dr. rer. nat) (Technical University Berlin. Metric Gradient Flows in Measure Spaces: Kernelization and Acceleration. Supervised by Prof. Gabriele Steidl. Referees: Prof. Anna Korba and Prof. José Antonio Carrillo de la Plata. Summa cum laude.)
04/2021 – 03/2023	Mathematics Master's (Technical University Berlin. Thesis: Wasserstein gradient flows - with an eye towards positive matrix-valued measures. Supervised by Prof. Gabriele Steidl and Dr. Robert Beinert.)
10/2017 – 03/2021	Mathematics Bachelor's (Technical University Berlin. Thesis: Atomic Norm Minimisation for Superresolution. Supervised by Prof. Gabriele Steidl and Dr. Robert Beinert.)

Positions

05/2026 – present	Postdoctoral Researcher (Department of Mathematics, Technical University of Munich. Chair: Massimo Fornasier; ERC Advanced Grant NEITALG)
04/2023 – 04/2026	PhD candidate (Applied Mathematics group at Technical University Berlin. Funding: stipend/BMBF project VI-Screen; TU Berlin teaching appointment from Jan. 2024.)
06/2021 – 03/2023	Student research assistant (Research on Wasserstein gradient flows, (re)writing lecture notes, proofreading manuscripts.)
10/2019 – 03/2021	Tutor (Department of Mathematics, Technical University Berlin.)

Affiliations

05/2026 – present	Junior Member (Munich Center for Machine Learning)
10/2025–04/2026	Phase II Student (Berlin Mathematical School)

Research visits

05/2026	One-week research visit, Heidelberg University (Host: Tim Laux. Topic: Geometry of accelerated dynamics on probability measures in (kernelized) Wasserstein geometries.)
11/2025	One-week research visit, University Carlos III Madrid (Host: Joaquín Míguez. Topic: Generative modeling with Bernstein approximations of f-divergences.)
08/2024 – 09/2024	One-month research visit, University of South Carolina (Host: Wuchen Li. Topic: Accelerated Stein metric gradient flows with general bilinear kernels on Gaussian families.)

Publications

01/2026	**V. Stein**, José Manuel de Frutos (equal contribution): Approximating f-Divergences with Rank Statistics (ICML'26. Proceedings of the 43rd International Conference on Machine Learning)
01/2026	**V. Stein**, Adwait Datar, Nihat Ay: Well-Posed KL-Regularized Control via Wasserstein and Kalman–Wasserstein KL Divergences (ICML'26. Proceedings of the 43rd International Conference on Machine Learning)
12/2025	J. Bresch, **V. Stein**: Interpolating between Optimal Transport and KL regularized Optimal Transport using Rényi Divergences (Results in Mathematics 81, 23 (2026))

07/2025	R. Duong, **V. Stein**, R. Beinert, J. Hertrich, G. Steidl: Wasserstein Gradient Flows of MMD Functionals with Distance Kernel and Cauchy Problems on Quantile Functions (ESAIM: Control, Optimisation and Calculus of Variations.)
06/2025	**V. Stein**, W. Li: Accelerated Stein Variational Gradient Flow (In: Nielsen, F., Barbaresco, F. (eds) Geometric Science of Information. GSI 2025. Lecture Notes in Computer Science, vol 16034. Springer, Cham, pp. 80–90.)
06/2025	R. Duong, N. Rux, **V. Stein**, G. Steidl: Wasserstein Gradient Flows of MMD Functionals with Distance Kernels under Sobolev Regularization (Philosophical Transactions of the Royal Society A, vol. 383, issue 0243 "Partial differential equations in data science".)
04/2025	**V. Stein**, S. Neumayer, N. Rux, G. Steidl: Wasserstein Gradient Flows for Moreau Envelopes of f-Divergences in Reproducing Kernel Hilbert Spaces (Analysis and Applications Vol. 24, No. 01, pp. 21-65 (2026).)

Preprints

03/2026	SympFormer: Accelerated attention blocks via Inertial Dynamics on Density Manifolds (with Wuchen Li, University of South Carolina, and Gabriele Steidl, Technical University Berlin.)
09/2025	Towards understanding Accelerated Stein Variational Gradient Flow - Analysis of Generalized Bilinear Kernels for Gaussian target distributions (with Wuchen Li, University of South Carolina.)

Conference and Workshop Talks

11/2026	SympFormer: Accelerated attention blocks via Inertial Dynamics on Density Manifolds (Minisymposium PDE, Measure Transport, and Geometric Perspectives on Generative Modeling, SIAM MDS in Utah, USA.)
11/2026	Wasserstein gradient flows of MMD-regularized f-divergences (Minisymposium Optimal Transport in Imaging – Flows, Regularization, and Machine Learning, SIAM IS in Utah, USA.)
09/2026	Wasserstein gradient flows of MMD-regularized f-divergences (BMS – BGSMath Junior Meeting 2026 in Barcelona, Spain.)
07/2026	Accelerated Stein Variational Gradient Flow (Minisymposium "New trends in geometric machine and deep learning" at Computational Methods in Applied Mathematics in Vienna, Austria, full DAAD-funding.)
06/2026	SympFormer: Accelerated attention blocks via Inertial Dynamics on Density Manifolds (Learning and Optimization in Luminy 2026)
03/2026	SympFormer: Accelerated attention blocks via Inertial Dynamics on Density Manifolds (MFO Workshop 2613 – Flows on Measure Spaces and Applications in Machine Learning in Oberwolfach, Germany.)
02/2026	What does Adam have to do with symplectic manifolds? The geometry behind momentum methods in machine learning (14th BMS Student Conference, Berlin.)
10/2025	Accelerated Stein Variational Gradient Flow (GSI'25: Geometric Structures of Statistical & Quantum Physics, Information Geometry, and Machine Learning (Saint-Malo, France), full DAAD-funding.)
09/2025	Wasserstein Gradient Flows for Moreau Envelopes of f-Divergences in Reproducing Kernel Hilbert Spaces (MML'25: Conference on Mathematics of Machine Learning 2025 in Hamburg, Germany.)

Seminar and Colloquium Talks

11/2025	Accelerated Stein Variational Gradient Flow (At the signal processing department colloquium of the University Carlos III Madrid.)
04/2025	Accelerated Stein Variational Gradient Flow (Stan Osher's level set online seminar.)
09/2024	Interpolating between Optimal Transport and KL regularized Optimal Transport using Rényi Divergences. (University of South Carolina Mathematics Graduate Colloquium)
08/2024	Wasserstein Gradient Flows of MMD Functionals with Distance Kernel and Cauchy Problems on Quantile Functions. (Joint Applied and Computational Mathematics (Changhui Tan & Siming He) and RTG data science seminar (Wuchen Li), University of South Carolina.)
08/2024	Wasserstein Gradient Flows for Moreau Envelopes of f-Divergences in Reproducing Kernel Hilbert Spaces (Stan Osher's level set online seminar.)

Poster Presentations

07/2026	FoCM 2026, Vienna. ("Wasserstein Gradient Flows for Moreau Envelopes of f-Divergences in Reproducing Kernel Hilbert Spaces", Computational Optimal Transport Session, "Accelerated Stein Variational Gradient Flow", Foundations of Data Science and Machine Learning session)
07/2026	ICML 2026, Seoul. ("Approximating f-Divergences with Rank Statistics" & "Well-Posed KL-Regularized Control via Wasserstein and Kalman–Wasserstein KL Divergences".)
03/2026	TUM's research retreat "Advanced Mathematical Methods in Data Analysis and Signal Processing", Tutzing, Germany. (Wasserstein Gradient Flows for Moreau Envelopes of f-Divergences in Reproducing Kernel Hilbert Spaces. Best Poster Award.)
02/2026	Gradient structures workshop in Augsburg, Germany. (Wasserstein Gradient Flows for Moreau Envelopes of f-Divergences in Reproducing Kernel Hilbert Spaces)
08/2025	Mathematical and Scientific Machine Learning, Naples, Italy. (Accelerated Stein Variational Gradient Flow)
10/2024	SIGMA (Signal - Image - Geometry - Modelling - Approximation) Workshop at the CIRM in France. (Wasserstein Gradient Flows for Moreau Envelopes of f-Divergences in Reproducing Kernel Hilbert Spaces)
06/2024	LOL: Learning and Optimization in Luminy, CIRM, France. (Wasserstein Gradient Flows for Moreau Envelopes of f-Divergences in Reproducing Kernel Hilbert Spaces)
03/2024	Workshop on Optimal transport from theory to applications, Berlin, Germany. (Wasserstein Gradient Flows for Moreau Envelopes of f-Divergences in Reproducing Kernel Hilbert Spaces)

Teaching at TU Berlin

Winter 2025/26	Harmonic Analysis (Lecture assistant)
Winter 2025/26	Numerical Mathematics I (Tutor)
Summer 2025	Mathematical Physics II - Statistical Mechanics (Lecture assistant)
Summer 2025	Probability Theory I (Tutor)
Winter 2024/25	Analysis II for Mathematicians (Tutor)
Winter 2024/25	Harmonic Analysis (Lecture assistant)
Summer 2024	Convex Analysis (Lecture assistant)
01/2024 - 02/2024	Numerical Mathematics I (Lecture assistant)

Supervised Thesis

03/2025 Roxane Leitheiser, Technical University Berlin (Wasserstein Gradient Flows of the MMD with Riesz Kernels on the Real Line. Bachelor's thesis. First supervisor: Gabriele Steidl.)

Service & Outreach

- Reviewer: JOTA, IPI, TMLR, IEEE/CAA Journal of Automatica Sinica, IEEE TNNLS, Mathematics, NeurIPS'24 workshop, program committee member NeurIPS'26 workshop.
- Reporter: Oberwolfach Report 26/13, Flows on Measure Spaces and Applications in Machine Learning.
- 2022/23: Tutored 14 seventh-grade students in weekly mathematical problem-solving sessions in Olympiad-style problems.
- 2022: Corrector, Tag der Mathematik team competition, Berlin.

Awards

Best Bachelor's Thesis Talk at the 17th annual Dies Mathematicus (TU Berlin, 2022).
Best Poster Award at the TUM's Research retreat "Advanced Mathematical Methods in Data Analysis and Signal Processing".

Technical Skills

Python, PyTorch, MATLAB, LaTeX; experience with HPC clusters.

Languages

German native; English fluent; French A2.

References

Prof. Dr. Gabriele Steidl
Institute of Mathematics,
Technical University of Berlin, Germany
steidl@math.tu-berlin.de

Prof. Wuchen Li
Department of Mathematics,
University of South Carolina, USA
wuchen@mailbox.sc.edu

Prof. Dr. Massimo Fornasier
Department of Mathematics,
Chair of Applied Numerical Analysis,
Technical University of Munich, Germany
massimo.fornasier@cit.tum.de

Prof. Dr. Tobias Breiten
Institute of Mathematics
Modeling, Simulation, & Optimization of Real Processes
Technical University Berlin, Germany
tobias.breiten@tu-berlin.de

Last update: August 2, 2026.